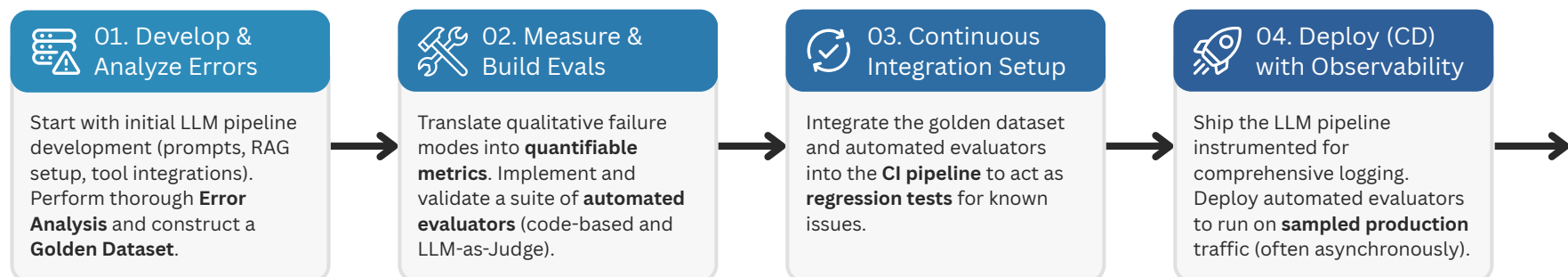


CI/CD for LLM Products:

The Continuous Improvement Flywheel



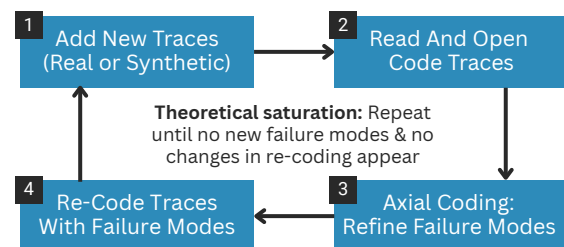
Start With Error Analysis & AI Evals



How to Perform Error Analysis?

The most successful teams look at data, identify failure modes, and let **app-specific metrics** emerge bottom-up.

You analyze **LLM traces** - **full records** of the pipeline: user query, input, reasoning, tool calls, and the output.



You need **~100** high-quality, diverse traces. Those can be real or synthetic data, or both coded with pass/fail and failure modes.

How to Build Automated Evals?

Once you perform initial error analysis, you can implement **automated evaluators** for single failure modes - each **evaluating a single metric**.

A. Start With Analyzing Failure Type

- | | |
|---|--|
| Specification Failure <ul style="list-style-type: none">Your instructions were unclear or incompleteFix the prompt first. Don't build an evaluator yet | Generalization Failure <ul style="list-style-type: none">LLM fails to apply clear instructions correctlyThese are candidates for automated evaluators |
|---|--|

B. Consider Two Types of App-Specific Evaluators

- | | |
|--|---|
| </> Code-Based Evals <ul style="list-style-type: none">Logic you write (e.g., .py)Objective, rule-based checks such as XM or RegexFast, cheap, deterministic | LLM-as-Judge Evals <ul style="list-style-type: none">Complex or subjective testsUses another LLM as judgeSingle, narrow failure modeStart with binary checks |
|--|---|

What is a Golden Dataset?

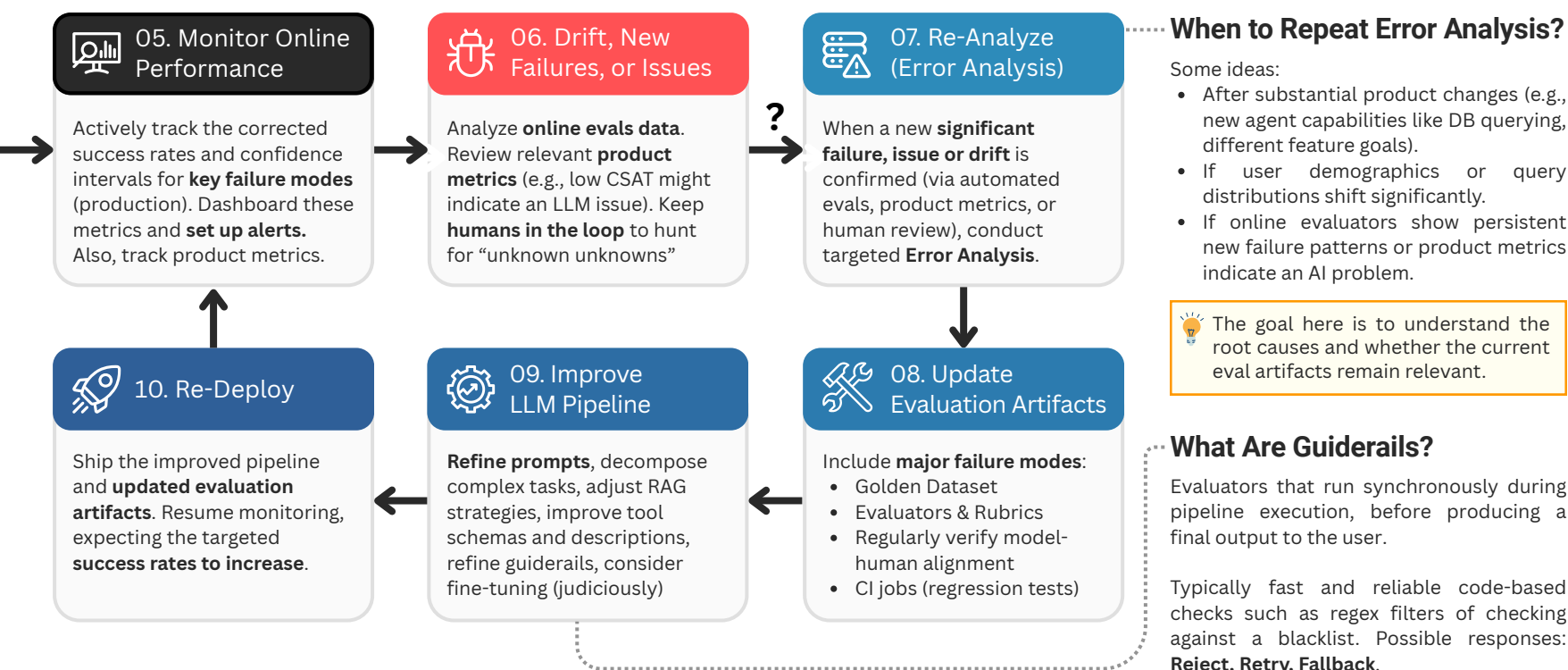
It's the foundation of LLM CI. A hand-curated set of input examples with corresponding reference outputs. Typically 100+ pairs.

On each proposed change we run all golden inputs through the pipeline and evaluate the outputs with a automated evaluators.

A Golden Dataset should:

- Cover the core features and capabilities
- Represent critical failure modes observed in the past
- Include edge cases
- Measure different elements of the pipeline (e.g., tools, RAG)

The Continuous Improvement Flywheel Continues



When to Repeat Error Analysis?

Some ideas:

- After substantial product changes (e.g., new agent capabilities like DB querying, different feature goals).
- If user demographics or query distributions shift significantly.
- If online evaluators show persistent new failure patterns or product metrics indicate an AI problem.

The goal here is to understand the root causes and whether the current eval artifacts remain relevant.

What Are Guiderails?

Evaluators that run synchronously during pipeline execution, before producing a final output to the user.

Typically fast and reliable code-based checks such as regex filters of checking against a blacklist. Possible responses: **Reject, Retry, Fallback**.